

**RED SEAL EXAM PREP**

# **310T Truck & Transport Mechanic**

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

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## Mock Exam — answer all questions, then check the key

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**Q1.** A diesel engine's EGR (Exhaust Gas Recirculation) valve is stuck in the open position. What symptoms would this cause at highway cruise speeds?

- A) Engine overheating — recirculated exhaust gases are very hot, increasing coolant temperature
- B) Turbocharger surge — excess flow from the EGR valve exceeds turbocharger capacity
- C) Rough running, loss of power, and black smoke as oxygen-depleted exhaust dilutes the intake charge
- D) No symptoms at cruise — EGR only operates at idle and low loads, not at highway speeds

**Q2.** What does "HPFP" stand for in a common-rail diesel fuel system, and what is its function?

- A) Hydraulic Pump Fuel Pressure — refers to fuel pressure at the hydraulic pump feed
- B) High-Performance Fuel Proportioner — balances fuel delivery to all cylinders
- C) High-Pressure Fuel Pump — pressurizes fuel to common rail pressure
- D) Hot-Phase Fuel Processor — heats fuel for cold-start performance

**Q3.** An engine uses a demand-regulated oil pump (variable displacement). The advantage over a fixed-displacement pump is:

- A) Lower parasitic loss — pumps only what is needed
- B) Provides more oil pressure at engine idle
- C) Simpler construction with fewer parts
- D) Eliminates the need for an oil pressure relief valve

**Q4.** When checking a driveshaft working angle, what is the maximum acceptable operating angle for most heavy truck driveshafts to prevent vibration and U-joint wear?

- A) 12 degrees
- B) 3 degrees
- C) 7 degrees
- D) 1 degree

**Q5.** A truck has multiple electrical accessories that stop working. The fuses are all good. What should the technician check NEXT?

- A) Check individual component grounds only
- B) Replace the BCM (Body Control Module)
- C) Check for a blown fusible link
- D) Replace the alternator

**Q6.** During a roadside inspection a truck with 10 service brakes is found with three brakes 6.4 mm (1/4 in) or more beyond the brake adjustment limit. Under the out-of-service criteria used at roadside in Canada, can the vehicle continue operating?

- A) Yes — up to 5 brake defects are allowed before out-of-service applies
- B) The 20% rule applies only to trailers, not tractor brake circuits
- C) No — 20% or more defective brakes puts the vehicle out of service
- D) Yes — only locked-up or missing brakes are cause for out-of-service

**Q7.** A trailer is equipped with an anti-lock braking system (ABS). After ABS repairs, the technician notices the ABS warning lamp stays on at all times during driving. What is the likely cause?

- A) The lamp is supposed to stay on — it only goes out during ABS modulation
- B) An active fault code remains — not cleared after repair or newly introduced
- C) Trailer ABS lamps are driven by the tractor ECU — it needs a software update
- D) The ABS is in bypass mode — the lamp shows normal non-ABS braking

**Q8.** A newly installed driveshaft vibrates at a frequency approximately twice per revolution of the shaft. What type of vibration is this and what is the most likely cause?

- A) Third-order vibration from an incorrect number of balance weights — three weights cause three pulses per revolution
- B) Second-order vibration from improper U-joint phasing or excessive operating angle
- C) Random vibration from loose driveshaft companion flange bolts — not frequency-correlated
- D) First-order vibration from driveshaft imbalance, which occurs once per revolution

**Q9.** Under NSC Standard 11, at a periodic inspection a truck's air brake system is rejected if, during the air leakage test (engine off, service brakes released), pressure drops faster than:

- A) 69 kPa (10 psi) per minute
- B) 41 kPa (6 psi) per minute
- C) 20 kPa (3 psi) per minute
- D) 7 kPa (1 psi) per minute

**Q10.** What is the MOST likely cause of excessive engine blow-by (excessive pressure/smoke from the crankcase breather)?

- A) Worn or broken piston rings
- B) Coolant leaking into the crankcase
- C) Faulty crankcase ventilation filter
- D) Engine oil overfilled

**Q11.** On a truck whose air compressor is gear-driven from the engine, what happens at the compressor when the governor reaches cut-out pressure?

- A) Governor air holds the inlet valves open and the compressor pumps unloaded
- B) A clutch in the compressor drive disengages and the compressor stops turning entirely
- C) The governor closes a valve in the discharge line so air cannot reach the reservoirs
- D) The safety valve on the supply reservoir lifts and vents the excess air to atmosphere

**Q12.** At a periodic commercial vehicle inspection under NSC Standard 11, a truck's windshield has a crack through one layer of glass that extends 30 mm into the area swept by the wipers. Under the standard, this windshield:

- A) Is rejected — any crack in the driver's swept area fails
- B) Passes — the crack is under 50 mm in the swept area
- C) Passes only if the crack is sealed before release
- D) Is rejected — the crack is longer than the 13 mm limit

**Q13.** A truck is in for a periodic commercial motor vehicle inspection under National Safety Code Standard 11. Every tire gauges within the recommended pressure and every tread depth is above the reject figure. Which of these remaining findings is a reject?

- A) A radial tire beside a non-radial on the same axle
- B) A steer tire built by a different maker than its mate
- C) Dual-mounted tires whose diameters differ by 8 mm
- D) A retreaded tire fitted to one of the drive axles

**Q14.** A high-mileage highway tractor with a non-synchronized 10-speed transmission pops out of gear into neutral while coasting down grades. The complaint occurs in one gear only. What is the most likely cause?

- A) Worn clutching teeth on that gear and its sliding clutch, tapered by repeated pop-out
- B) Loose transmission mounts letting driveline vibration shake the gear out of engagement
- C) A worn blocker ring in the synchronizer for that gear letting the collar back out
- D) A slipping clutch, which pushes the transmission out of gear when the driver lifts off

**Q15.** During a pre-trip air brake test, a driver finds the air pressure drops more than 3 PSI per minute with the engine off and brakes released. What does this indicate?

- A) Excessive system air leakage that must be located and repaired
- B) Normal — 3 PSI per minute is within acceptable leakage limits
- C) The air dryer purge valve is open — close it to stop the leak
- D) The governor is faulty — it should hold pressure with the engine off

**Q16.** A truck clutch is slipping — engine RPM rises sharply during acceleration but vehicle speed does not increase proportionally. What would cause clutch slippage?

- A) Clutch disc too thick — excess disc material increases friction surface contact
- B) Oil or grease contamination on the clutch disc friction surfaces from a leaking rear main seal
- C) Excessive clutch free travel — too much free play causes the pressure plate to apply too much force
- D) Incorrect flywheel surface roughness — a smooth flywheel increases friction and causes slippage

**Q17.** Following a foundation brake overhaul, a technician sets the initial adjustment on a newly fitted automatic slack adjuster by the manufacturer's installation procedure, then backs it off to confirm lining-to-drum clearance. The adjuster has not taken up on the first four or five brake applications. What is the correct assessment?

- A) Normal — ASAs adjust gradually over multiple applications
- B) The ASA pawl is broken — it should adjust on the first application
- C) The brake linings are worn out and preventing proper adjustment
- D) The adjuster is defective and must be replaced right away

**Q18.** A carrier's safety manual cites NSC Standard 13 as the authority for its post-trip inspection policy. Which statement about post-trip inspections in the National Safety Code is correct?

- A) Post-trip inspection appears nowhere in the National Safety Code at all
- B) Standard 14 sets the post-trip interval used in the carrier safety rating
- C) Standard 13 requires a post-trip inspection report at the end of each day
- D) Standard 16 makes post-trip inspection a mandatory training competency

**Q19.** A driver discovers a major defect in the brakes during the daily trip inspection — the pushrod stroke is beyond maximum on three wheels. What must the driver do?

- A) Repair the brake adjustment himself before departure — drivers can perform minor brake adjustments to correct out-of-adjustment brakes
- B) Not operate the vehicle — report to the carrier, record the defect, and repair before operation
- C) Drive to the nearest repair facility — if the brakes still function, a major defect can be addressed after completing the current trip
- D) Record the defect at the end of the day — the report only covers defects found before the trip began

**Q20.** After coupling a tractor to a semi-trailer, which check confirms that the fifth wheel is properly locked to the trailer kingpin?

- A) A tug test plus visual confirmation that the jaws are locked around the kingpin with no coupler gap
- B) Raising the landing gear fully and confirming the trailer follows the tractor when driving forward
- C) Checking that the fifth wheel plate is well greased and the kingpin shows witness marks
- D) A tug test alone — if the tractor cannot pull out from under the trailer, the coupling is complete

**Q21.** A truck's trailer light circuit has no tail lights on the trailer. The tractor tail lights work normally. The first check should be:

- A) Replace the trailer light module with a known-good unit
- B) Test the trailer harness from front to rear with a wiring tracer
- C) Check the 7-pin connector for corrosion or damaged pins
- D) Check the trailer ABS module — it controls all trailer lights

**Q22.** What is the purpose of "pilot injection" in a modern common-rail diesel injection system?

- A) To inject a small charge before the main event, reducing knock
- B) To pre-fill the injector needle for faster main injection response
- C) To inject fuel into the exhaust stream for DPF regeneration
- D) To test injector function during each cycle before main injection

**Q23.** A truck's clutch pedal free travel is too excessive (more than specified). What is the effect on clutch operation?

- A) Clutch chatter during engagement — excessive free travel destabilizes the clutch disc
- B) No effect — clutch free travel has no impact on clutch engagement or disengagement
- C) The clutch may not fully disengage, making gear changes difficult and causing gear grinding
- D) Clutch slips under load — excessive free travel means the pressure plate remains applied too lightly

**Q24.** A truck's alternator is producing 14.2V but the battery is not fully charging after long trips. What should be suspected?

- A) The alternator voltage is too high — it is overcharging the battery
- B) The voltage regulator is stuck in the low-voltage position
- C) High resistance in the charging circuit or a failing battery
- D) The battery needs distilled water — low electrolyte prevents charging

**Q25.** A late-model diesel engine logs a fault for "NOx sensor downstream of SCR catalyst reading higher than expected." What is the MOST likely cause?

- A) DEF quality or concentration is out of spec, cutting SCR efficiency
- B) The DPF is loaded — soot backpressure is causing incomplete combustion
- C) The NOx sensor itself is failed high — replace sensor immediately
- D) The EGR valve is stuck open, routing excess exhaust into the intake

**Q26.** What voltage is standard for most North American heavy truck electrical systems?

- A) 24V
- B) 48V
- C) 6V
- D) 12V

**Q27.** A loaded truck scales 22,000 kg on its tandem drive axle. For that axle group the limit is 17,000 kg under the federal-provincial-territorial Memorandum of Understanding on interprovincial weights and dimensions, which each province enacts in its own highway legislation. The driver is stopped at a weigh scale. What are the possible consequences?

- A) The truck is seized on the spot, because overweight violations always mean seizure
- B) The driver is given a warning, because a first overweight offence carries no fine
- C) Only the carrier is fined, because the driver is not responsible for the load weight
- D) A fine for the driver, possible charges against the carrier, and an order to offload

**Q28.** Under Canadian regulations, what is required when transporting dangerous goods (DG) by road?

- A) A dangerous goods endorsement added to the driver's licence
- B) Training, shipping documents, and vehicle placarding
- C) Shipping documents and placards, but training only for the shipper
- D) Placards on the vehicle and a training certificate, but no shipping document

**Q29.** A heavy truck carries four 12-volt batteries wired in parallel. One is replaced with a new battery of the same type while the other three stay in service. What is the usual consequence?

- A) The bank still performs at the level of its weakest battery
- B) The bank now supplies 48 volts because the batteries are in parallel
- C) The alternator will overcharge because bank resistance has dropped
- D) The new battery raises the whole bank back to full cranking output

**Q30.** A truck diesel engine has low oil pressure at idle but normal pressure at high RPM. The MOST likely cause is:

- A) Worn main or rod bearings
- B) Blocked oil filter
- C) Oil viscosity too high
- D) Faulty oil pressure sending unit

**Q31.** An air brake system governor is set at 125 PSI cut-out and 100 PSI cut-in. A technician resets the governor so that cut-out occurs at 175 PSI. What is the effect on system components?

- A) Pressure beyond what the dryer, valves, fittings and reservoirs are rated for
- B) No change — every component in an air brake system is rated to at least 175 PSI
- C) Stronger parking brake hold — the spring brakes clamp harder at higher pressure
- D) The compressor cycles more often because cut-out now sits further from cut-in

**Q32.** A truck body module controls the exterior lighting with PWM outputs. The headlights dim randomly. A DMM on the headlight feed shows a steady average voltage that does not change when the lights dim. What tool would best diagnose the intermittent signal fault?

- A) DMM set to Hz (frequency) mode
- B) A 12-volt incandescent test light
- C) An oscilloscope on the PWM output
- D) Amp clamp around the headlight wire

**Q33.** What is the function of a driveshaft's slip yoke?

- A) To allow the driveshaft to rotate at different speeds than the transmission output when cornering
- B) To absorb torque spikes from the engine and protect the transmission output shaft
- C) To allow quick driveshaft removal for service without disconnecting universal joints
- D) To allow the driveshaft to change length as the suspension moves the axle relative to the frame

**Q34.** What is the purpose of the "dither" signal in a proportional solenoid control system for an electronically controlled transmission?

- A) Dither is a rapid alternating polarity signal that drives the solenoid in both directions to ensure it cannot stick
- B) A high-frequency, low-amplitude oscillation superimposed on the command signal to prevent solenoid stiction
- C) Dither refers to the controller's delay time between receiving a command and activating the solenoid
- D) Dither is a diagnostic mode that cycles solenoids to test for sticking during pre-operation checks

**Q35.** A truck driver reports a vibration that occurs only at highway speeds (above 90 km/h) and goes away below 70 km/h. The vibration is felt through the floor, not the steering wheel. What is the most likely cause?

- A) Engine misfire — engine misfires create vibration through the body at high RPM only
- B) Driveshaft imbalance or phasing error — felt through the floor within a specific speed range
- C) Front wheel imbalance — imbalance is always felt through the steering wheel
- D) Tire flat spot from hard braking — flat spots occur at any speed, not only above 90 km/h

**Q36.** A truck diesel engine cranks normally but will not start in cold weather. The intake air pre-heater (grid heater) has been verified as working. What should be checked NEXT?

- A) Check the ECM calibration level
- B) Check injector return (spill) volume
- C) Check fuel quality and filter
- D) Check the engine valve timing

**Q37.** What does "rated RPM" mean for a truck diesel engine?

- A) The idle RPM programmed into the ECM for fuel economy
- B) The governor cut-off RPM that prevents engine over-speed
- C) The RPM at which the engine produces maximum torque
- D) The full-load speed at which maximum power is made

**Q38.** A relay valve in an air brake system is located at the rear axle brakes. Its purpose is to:

- A) Speed brake application using air from a local reservoir
- B) Automatically adjust rear brake pressure based on load
- C) Reduce maximum brake pressure to the rear axle
- D) Prevent brake pressure from exceeding governor cut-out pressure

**Q39.** What is the purpose of the two-way check valve (double check valve) in an air brake system?

- A) Allows air to flow in both directions at once for dual-circuit charging
- B) Selects the higher of two inlet pressures and passes it to the outlet
- C) Prevents air flowing backward from the reservoirs to the compressor
- D) Reduces primary-to-secondary pressure to balance brake application

**Q40.** A truck clutch requires adjustment because the free travel has decreased to near zero. If ignored, what will result?

- A) Harsh and noisy gear engagement on every shift
- B) Increased effort needed at the clutch pedal
- C) Clutch slipping from release bearing contact
- D) Difficulty disengaging the clutch for shifting

**Q41.** Trucks, tractors and trailers must be inspected every 24 hours and the driver completes a trip inspection report (some carriers call it a vehicle condition report). Under NSC Standard 13, the carrier must retain that report for a minimum of:

- A) 30 days
- B) 1 year
- C) 6 months
- D) 7 days

**Q42.** What does it mean when a diesel engine's oil pressure gauge reads zero while the engine is running at normal operating temperature?

- A) An emergency — shut down the engine immediately to prevent damage
- B) The oil is hot and thin — pressure drops to zero at operating temperature
- C) This is normal at idle — oil pressure only registers above 1,000 RPM
- D) The gauge sender has failed — keep operating and replace it at next service

**Q43.** A Class 8 highway tractor has a bobtail proportioning system in its rear-axle service brake circuit. When the tractor runs bobtail (with no trailer coupled), what does it do?

- A) It sends the unused trailer-circuit air to boost the tractor reservoirs
- B) It raises rear brake pressure to make up for the missing trailer brakes
- C) It reduces drive-axle brake pressure because the axle is lightly loaded
- D) It shuts the rear service brakes off so the steer axle does the stopping

**Q44.** A Class 8 tractor-trailer fails a pre-trip inspection because the spring brake parking function will not hold the vehicle stationary on a grade. The reservoirs are full and the pushrod strokes are inside the adjustment limit. Which component failure MOST likely explains this?

- A) A broken power spring in a spring brake chamber, which cannot develop full clamping force
- B) The tractor protection valve is closed, cutting off air to the trailer
- C) The service brake relay valve is not releasing service air from the chambers
- D) The low air pressure warning switch has failed, so the spring brakes were never signalled to apply

**Q45.** A truck has 4 inches of free play at the steering wheel rim before the front wheels begin to move. What does the National Safety Code use as the limit for steering wheel free play?

- A) No limit at all — free play is not part of a safety inspection
- B) A flat 2 inches of free play at the rim on every commercial vehicle
- C) A flat 1 inch of free play at the rim on every commercial vehicle
- D) Limits that vary with wheel diameter and manual or power steering

**Q46.** A drive axle is fitted with a limited-slip differential. What does its oil need that a plain differential's oil does not?

- A) A detergent package to keep the clutches clean
- B) A higher viscosity oil than the plain axle uses
- C) A GL-4 rating instead of the usual GL-5 rating
- D) A friction modifier additive for the clutch pack

**Q47.** A truck's battery isolator switch is turned off overnight but the battery is still discharged by morning. The MOST likely cause is:

- A) Normal self-discharge of the battery over 8 hours
- B) The starter motor is drawing current continuously
- C) The isolator switch is not fully isolating the battery
- D) The alternator is draining the battery when engine is off

**Q48.** What is the function of the quick-release valve in an air brake system?

- A) Quickly increases brake application pressure for emergency stops
- B) Allows air to bypass the chambers when the parking brake is engaged
- C) Regulates maximum brake application pressure to prevent lockup
- D) Rapidly exhausts chamber air when the brakes are released

**Q49.** A circuit protected by a 10-amp fuse keeps blowing the fuse. The CORRECT diagnostic approach is:

- A) Install a 20-amp fuse to stop the blowing
- B) Add a second 10-amp fuse in parallel
- C) Replace the fuse with a circuit breaker of any rating
- D) Find and repair the cause of the overcurrent

**Q50.** An engine oil cooler bypass valve opens when:

- A) Oil is cold and the pressure differential is high
- B) The oil pump output exceeds system pressure
- C) Oil temperature exceeds the maximum limit
- D) Coolant temperature exceeds 95°C

## Answer Key & Explanations

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### Q1. Answer: C

**EGR stuck open at cruise: rough running, loss of power, excessive exhaust smoke.** At highway speeds, the engine needs maximum fresh air for full-load combustion. EGR gases are depleted of oxygen — introducing them at cruise reduces the effective air charge. The combustion is oxygen-starved, causing rich combustion (black smoke), loss of power, and rough running. The EGR is typically commanded closed at high load/high RPM by the ECM — a stuck-open mechanical valve overrides this. The engine may also misfire.

### Q2. Answer: C

**HPFP = High-Pressure Fuel Pump.** In a common-rail system, the HPFP raises fuel pressure from transfer pump pressure (~5–10 bar) to injection pressure (typically 1,600–2,200 bar / 23,000–32,000 PSI). This high-pressure fuel is stored in the common rail and distributed to solenoid or piezo injectors. The HPFP is driven by the engine and is the heart of the common-rail injection system.

### Q3. Answer: A

**Variable displacement oil pump: matches output to demand — less heat and better fuel economy.** A fixed pump always operates at maximum displacement, burning extra power circulating oil through the relief valve. A demand-regulated pump reduces displacement at highway speeds (lower demand), saving 1–3% fuel. It still maintains required oil pressure and flow at all operating conditions.

### Q4. Answer: B

**Maximum continuous operating angle: approximately 3 degrees.** U-joints run at large angles generate torsional excitation and wear rapidly; higher angles are tolerated only at lower shaft speeds. Two companion rules: the operating angles at the two ends of the shaft must be equal within 1 degree, and every joint needs at least 1 degree of angle so its needle rollers keep rotating instead of brinelling the cross. Cancellation of the speed fluctuation requires BOTH conditions met at once - equal angles at the two joints AND correct phasing, meaning the in-board yokes of the shaft in line with each other. A shaft that is out of phase, or whose joint angles are unequal, leaves the second joint adding to the first joint's fluctuation instead of cancelling it. Check phasing and measure the angles as separate steps of a driveline inspection, and remember ride height changes the angles.

### Q5. Answer: C

**Multiple circuits dead with good fuses = upstream fault.** A blown fusible link or tripped main circuit breaker affects all circuits downstream. Fusible links are located near the battery and protect main wiring harnesses — they look like wire but contain a sacrificial element.

**Q6. Answer: C**

**Three defective brakes out of ten is 30% — at or over the 20% threshold, so the vehicle goes out of service.** The North American Standard Out-of-Service Criteria, administered by the Commercial Vehicle Safety Alliance and applied by Canadian enforcement at roadside, place a vehicle out of service when the number of defective brakes is equal to or greater than 20% of the service brakes on the vehicle or combination. A brake 6.4 mm (1/4 in) or more beyond its adjustment limit counts as one defective brake; a brake found exactly at the limit is not counted at all. The criteria's own chart shows that two defective brakes are enough to declare a ten-brake vehicle out of service, so three is well over. NSC Standard 11 notes that these roadside criteria are similar but not identical to its own shop-based Hazardous Conditions.

**Q7. Answer: B**

**ABS warning lamp on during driving = active ABS fault code.** Lamp sequence: illuminates at key-on for bulb check, then extinguishes after a short self-test if no faults. Continuous illumination during operation indicates an unresolved stored fault. After ABS repairs, fault codes must be cleared using a service tool or diagnostic software. If the lamp re-illuminates, a new fault was introduced during the repair (wiring disturbed, sensor reconnected incorrectly) or the existing fault was not actually repaired.

**Q8. Answer: B**

**Vibration twice per revolution is second order, and it comes from U-joint phasing or an excessive operating angle.** A single U-joint operating at any angle creates velocity variation — the output shaft alternately speeds up and slows down twice per input revolution, which is second order. Two correctly phased U-joints cancel that variation. Incorrect phasing, meaning the yokes at the two ends are not in the same plane, makes the second variation add instead of cancel and produces a strong twice-per-revolution vibration. Excessive operating angle at either joint increases the magnitude of the variation as well, so a shaft can be correctly phased and still shake if the angles are large or unequal. Because the shaft is newly installed, the first thing to check is whether it went back together in the orientation it came apart in.

**Q9. Answer: D**

**Reject at more than 7 kPa (1 psi) per minute with the brakes released.** NSC Standard 11 (January 2020), Part B Section 3A, item 2 'Air Supply System', condition (g), rejects a truck whose air pressure drops at more than 7 kPa (1 psi) per minute during the leakage test. A drop of more than 20 kPa (3 psi) per minute is that section's hazardous-condition (out-of-service) tier, not the acceptance limit. Condition (f) of the same item separately rejects a loss of more than 138 kPa (20 psi) during a full service application. The driver-level applied test uses different figures: with the brakes held fully applied, the Ontario Official Air Brake Handbook allows 21 kPa (3 psi) per minute for a straight truck, tractor or bus, 28 kPa (4 psi) per minute for a tractor-trailer, and 41 kPa (6 psi) per minute for a tractor with two or more trailers.

**Q10. Answer: A**

**Excessive blow-by = combustion gases past piston rings.** All engines have some blow-by — rings can't achieve a perfect seal. Worn or broken rings allow significantly more combustion gas into the crankcase, pressurizing it. Signs: oil pushed out seals and breathers, smoky breather outlet, rapid oil contamination. Measure with a blow-by flow meter to confirm.

**Q11. Answer: A**

**At cut-out the governor pilots reservoir air down the unloader line to the compressor, and that air holds the compressor's inlet valves open. The compressor keeps turning, but it simply shuttles air back and forth instead of pumping it into the reservoirs.** A truck air compressor is driven off the engine gear train (or a belt) and turns the whole time the engine runs, so nothing disengages it and there is no shutoff valve in the discharge line. When reservoir pressure falls to cut-in, the governor exhausts the unloader line, the inlet valves close, and the compressor loads again. The reservoir safety valve is a last-resort pop-off, not the normal pressure control - if it is lifting, something in the governor or unloader circuit has already failed. This is why a compressor that will not stop building, or one that never builds at all, is diagnosed by testing the governor and the small line running to the compressor unloader before the compressor itself is condemned.

**Q12. Answer: B**

**Reject only when the damage passes the stated limits.** NSC Standard 11 rejects a windshield when a crack extends through both layers of glass, when a crack of any length extends more than 50 mm within the area swept by the OEM wipers, or when a chip larger than 13 mm in diameter lies in that area. The standard prints the matching pass conditions, including a crack that extends less than 50 mm into the swept area, and a crack longer than 50 mm that lies outside the swept area. The 13 mm limit belongs to chips, not to cracks. Note also that a crack in the driver's line of sight does not automatically stop the vehicle: on the daily trip inspection under Standard 13, glass that fails to give the driver the required view is listed as a defect and not as a major defect, so it is recorded and reported and the vehicle may continue in service pending repair.

**Q13. Answer: A**

**A tire inspection covers four things - depth, condition, inflation and matching - and mixing a radial with a non-radial on one axle fails on the matching criteria.** The tire criteria for a periodic inspection sit in National Safety Code Standard 11, Part B, Section 9 - Tires and Wheels, and they run across four items: tread depth, tread condition, tire sidewall and manufacturer markings, and tire inflation pressure. Matching belongs to the sidewall and markings item, under matching and application, which rejects when the nominal tire size difference on an axle is greater than 25 mm, when dual-mounted tire diameters differ by more than 13 mm, when wheel or rim size does not match tire size, when a required tire is missing, when a radial tire is mixed with a non-radial on an axle, or when a tire is labelled in a way that shows it is not intended for on-road use. Bias and radial tires on one axle are listed as a hazardous condition as well, so that truck is not fit to drive away either. The other three findings are not criteria: duals 8 mm apart sit inside the 13 mm limit, a retreaded tire is barred from an active steering axle rather than from a drive axle, and nothing in the section turns on who made the tire or how old it is - matching is judged by size, construction and diameter. Correct pressure and legal tread satisfy only two of the four things a tire inspection has to cover.

**Q14. Answer: A**

**One gear only, and only on the overrun: the clutching teeth for that gear are worn.** In a non-synchronized heavy-duty box the sliding clutch is held on the gear by its clutching (dog) teeth. Repeated pop-out tapers those teeth, and once tapered they act like a ramp: the reversed driveline load of coasting drives the sliding clutch back off the gear, which is why the fault shows on deceleration and in that gear alone. A blocker ring cannot be the cause, because a non-synchronized main section has no synchronizers to wear. Loose mounts and a slipping clutch act on the whole drivetrain and would not single out one gear. Read the pattern first: a fault confined to one gear points at that gear pair, while a fault in every gear points at the shift rails, detents or linkage.

**Q15. Answer: A**

**More than 3 PSI per minute with the brakes released is excessive leakage under every Canadian standard — find it and repair it.** On periodic inspection, NSC Standard 11 Part B Section 3A rejects a truck whose air pressure drops more than 7 kPa (1 psi) per minute during the leakage test, and lists a drop of more than 20 kPa (3 psi) per minute as a Hazardous Condition; a trailer is rejected above 28 kPa (4 psi) per minute when attached to a towing vehicle, or above 20 kPa (3 psi) per minute on a shop air source, and is hazardous above 40 kPa (6 psi) per minute. The driver-level test in the Ontario Air Brake Handbook is done with the engine off and the service brakes fully applied, allowing 21 kPa (3 psi) per minute for a straight truck, tractor or bus, 28 kPa (4 psi) for a tractor and trailer, and 41 kPa (6 psi) for a tractor with two or more trailers. Leakage with the brakes released should be lower than with them applied, so this reading fails all of them. Build to governor cut-out, shut the engine off, time the drop, then hunt the leak with soapy water.

**Q16. Answer: B**

**Clutch slipping: oil/grease contamination on friction surface — pressure plate cannot grip the disc.** Oil from a leaking rear main seal or improperly lubricated parts contaminates the disc friction material. Oil-soaked friction material has greatly reduced coefficient of friction. Other causes: worn friction disc (too thin, past minimum thickness), weak or broken pressure plate springs, overheated pressure plate (springs lose tension from clutch abuse). Contamination: replace both disc and pressure plate — cleaning oil from friction material is ineffective.

**Q17. Answer: A**

**Normal in this context: an automatic adjuster takes up clearance a little at a time over a series of applications, not in one.** The adjuster only advances when the stroke reaches its adjustment threshold, so after a deliberate back-off during installation it may need several full applications at operating pressure before it has ratcheted back to the correct clearance. Run the number of applications the manufacturer's procedure calls for, then measure the stroke. Do not condemn a new unit for failing to recover on the first pull, and do not read the delay as a broken pawl. Two limits sit on this answer. First, it describes the installation and verification procedure, where backing off is part of the published method — it is not a description of what to do to a truck in service. Second, if the stroke is still outside the limit once the procedure is complete, the unit is not slow, it is faulty, and the correct response is repair or replacement of the adjuster or of whatever in the foundation brake is defeating it. That distinction matters because an automatic adjuster found at or beyond its stroke limit in service must never be wound back by hand to make it pass: National Safety Code Standard 11 states that such a brake requires repairs and that a manual adjustment will not correct the problem.

**Q18. Answer: D**

**Standard 13 never uses the term; Standard 16 does, and marks it mandatory.** National Safety Code Standard 13 is the trip inspection standard, and the words post-trip and pre-trip do not appear anywhere in it. What it requires is a Schedule 1 inspection every 24 hours, an inspection report that records the fact where no defects are found, and a duty on the driver to monitor the vehicle while driving, record any defect detected on the inspection report and report it to the carrier before the next required inspection — immediately if the defect is a major one, in which case the vehicle is not to be driven. Post-trip appears instead in Standard 16, Entry Level Training, where competency 7.2.11, 'Conducts regular enroute and post-trip vehicle inspections', is marked M for mandatory; items marked R for recommended sit in the same table, so the marking is unambiguous. Standard 15, the facility audit standard, has a section headed Pre/Post Trip Inspections and defines a Pre/Post Trip Inspection violation as an audited finding against the carrier. Standard 14, Safety Rating, contains no occurrence of the term at all. The carrier's policy is real enough; Standard 13 is simply the wrong authority to hang it on.

**Q19. Answer: B**

**Major defect found at the daily trip inspection: the vehicle must NOT be operated until it is repaired and certified.** National Safety Code Standard 13 splits every schedule into a defect column and a major defect column, and the two carry different duties. Schedule 1 lists pushrod stroke of any brake exceeding the adjustment limit as a major defect, so one wheel over stroke already qualifies and three wheels is far past it. Standard 13 then states that no motor carrier shall permit a person and no person shall drive a commercial vehicle on a highway when a major defect is present. The driver records the defect on the inspection report, does not operate the vehicle, and reports the condition to the carrier immediately. The carrier has the defect repaired and certified before the vehicle returns to service. A driver may not correct this by adjusting the brakes at the roadside, and there is no allowance for finishing the current trip first.

**Q20. Answer: A**

**Coupling is not complete without a visual inspection — a tug test alone can pass on a false (high) couple.** Ease forward with trailer brakes applied to feel resistance, then secure the vehicle and physically look: locking jaws fully closed around the kingpin shank (not on top of or ahead of it), no visible gap between the trailer upper coupler plate and the fifth wheel surface, and the release handle fully seated in the locked position. A kingpin riding on top of the fifth wheel plate or caught ahead of the jaws can hold during a tug test and then drop the trailer on the road.

**Q21. Answer: C**

**Trailer lights out: start at the 7-pin connector, and check the trailer circuit fuse.** The 7-way connector is the most common cause of trailer lighting issues. Moisture, corrosion, and physical damage to pins cause high resistance or open circuits. Clean with electrical contact cleaner, inspect pins for spread contacts or corrosion. Also check the tractor-side fuse for the trailer circuit.

**Q22. Answer: A**

**Pilot injection: small pre-injection reduces diesel knock.** A very small amount of fuel injected slightly before the main event ignites first, raising cylinder temperature and reducing the ignition delay of the main charge. Diesel knock is caused by the pressure spike from rapid combustion of fuel that accumulated during ignition delay. Result: more gradual pressure rise, quieter and smoother combustion, less stress on bearings and pistons. Modern systems may use 3–5 injection events per cycle.

**Q23. Answer: C**

**Excessive clutch free travel = insufficient pedal travel remaining to fully disengage clutch.** Free travel is the pedal movement before the throwout bearing contacts the release fingers. After free travel is used up, the remaining pedal stroke releases the clutch. If free travel is too large, there isn't enough remaining pedal travel to fully disengage — the clutch drags (partially engaged) during shifting, causing gear grinding and premature synchronizer wear.

**Q24. Answer: C**

**Correct alternator voltage but poor charging: check current flow and circuit resistance.** 14.2V is within spec, but if the charging circuit has high resistance (corroded battery cable or charge wire, poor battery terminal connection), current flow to the battery is restricted despite correct voltage output. A battery with high internal resistance also won't accept charge efficiently. Test: measure charging current (clamp ammeter), measure voltage drop in the charging circuit, load test the battery.

**Q25. Answer: A**

**DEF quality and concentration are the first thing to suspect when SCR conversion efficiency drops.** Selective catalytic reduction uses diesel exhaust fluid, a 32.5 percent urea solution, to convert NOx to nitrogen and water. Diluted, contaminated, or frozen-then-improperly-thawed fluid degrades that conversion, and the downstream NOx sensor sees the result. The same fault family is triggered by low fluid level, a dosing injector fault, a fluid heater fault, and a degraded catalyst. A loaded particulate filter does not raise downstream NOx, and an EGR valve stuck open lowers NOx rather than raising it. Verify fluid quality and level before condemning the sensor or the catalyst.

**Q26. Answer: D**

**12V is standard for most North American trucks.** European trucks commonly use 24V. Some newer hybrid systems use 48V. Two 12V batteries wired in parallel = 12V (more capacity). Two wired in series = 24V (used in European and some heavy equipment).

**Q27. Answer: D**

**Overweight consequences: a fine for the driver, possible charges against the carrier, an order to reduce the load, and a mark on the carrier's safety record.** Overweight fines are set provincially and rise with the amount over the limit. Depending on the circumstances the driver, the carrier and sometimes the shipper can all be charged. The vehicle can be held at the scale until the excess is removed or the load is redistributed within the axle limits. Repeated overweight convictions show up on the carrier's Commercial Vehicle Operator's Registration record in Ontario, and on the equivalent carrier profile in other jurisdictions, and feed into the safety rating.

**Q28. Answer: B**

**The Transportation of Dangerous Goods Act, 1992 and its Regulations set three duties that travel together.** Anyone who handles, offers for transport or transports dangerous goods must be adequately trained and hold a valid training certificate, or do the work in the presence and under the direct supervision of someone who is — so training does not stop at the shipper. A shipping document must accompany the shipment, and the vehicle must display the correct safety marks for the class and quantity being carried. An Emergency Response Assistance Plan is required for certain higher-risk goods. There is no dangerous goods endorsement on a Canadian driver's licence; the training certificate is the credential.

**Q29. Answer: A**

**Parallel batteries share one terminal voltage, so the bank is only as good as its worst battery.** Wiring in parallel holds the system at 12 volts and adds cranking capacity, but it also ties every battery to the same voltage. A tired battery sitting beside a new one becomes a load on it: the new battery pushes current into the old one instead of into the starter, and it sits at a partial state of charge that shortens its own life. This is why heavy truck batteries are replaced as a matched set of the same type, rating and age, and why the starter feed and the ground are taken from opposite corners of the bank so current has an equal path through each battery. Series wiring is what raises voltage — parallel wiring never does.

**Q30. Answer: A**

**Pressure low at idle, OK at RPM = worn bearings.** Worn main or rod bearings with excessive clearance allow oil to escape faster than the pump supplies at low RPM; at high RPM, pump output overcomes the leakage. Compare oil pressure to spec: typically 10 PSI per 1000 RPM minimum.

**Q31. Answer: A**

**A 175 PSI cut-out drives the supply side past what the dryer, valves, fittings and reservoirs are built to hold.** The Official Air Brake Handbook sets the ceiling plainly: actual cut-out pressure must never be higher than 1000 kPa (145 psi), and cut-out normally sits only 138 to 173 kPa (20 to 25 psi) above cut-in. Safety valves normally open at 1035 kPa (150 psi), so a 175 PSI setting leaves the reservoir safety valve venting continuously — and a safety valve that is venting air means too much pressure is building in the system and needs immediate repair. Seals, hoses, the dryer purge valve and fittings all work above their design pressure, and the compressor runs loaded far longer than intended. The extra pressure buys nothing at the brakes: application pressure is metered by the treadle and relay valves, and the spring brakes are held off by air but applied by their springs, so parking hold does not change. Reset the governor to the OEM cut-in and cut-out figures with a calibrated gauge.

**Q32. Answer: C**

**Oscilloscope: captures the waveform over time — reveals intermittent faults.** A DMM reads average/RMS voltage and misses brief dropouts. An oscilloscope displays the actual PWM waveform — a dropout, missing pulse, or duty cycle change appears instantly on screen. Essential for intermittent electronic faults on PWM-controlled circuits.

**Q33. Answer: D**

**Slip yoke: allows driveshaft length change as suspension cycles.** When the rear axle moves up and down over bumps, the distance between the transmission output and the axle pinion changes. The slip yoke telescopes on the output shaft splines to accommodate this length change. Without a slip yoke, suspension movement would either stretch or compress the driveshaft, causing severe vibration or binding. The slip yoke splines must be lubricated to prevent wear and noise.

**Q34. Answer: B**

**Dither: high-frequency low-amplitude signal superimposed on command to prevent solenoid stiction.** Without dither, a proportional solenoid may stick at its current position due to static friction (stiction) between the plunger and bore. This causes hysteresis — the output at a given command differs depending on whether command is increasing or decreasing. Dither (typically 100–500 Hz) keeps the plunger in continuous slight motion, breaking static friction and ensuring the plunger position precisely tracks the command signal.

**Q35. Answer: B**

**Speed-specific floor/seat vibration = likely driveshaft balance or phasing issue.** Driveshaft vibration is proportional to rotational speed (square of speed for imbalance). At low speeds, even significant imbalance produces tolerable vibration. Above a threshold speed, vibration becomes severe. A phasing error — the yokes at each end of the shaft not aligned in the same plane — leaves the two joints' speed fluctuations adding instead of cancelling, which creates a second-order vibration. Front imbalance vibrates through the steering column/wheel. Rear driveshaft vibration transmits through the cab floor. Confirm with a driveshaft balance check and universal joint phasing inspection.

**Q36. Answer: C**

**Cold no-start with a working intake pre-heater points at the fuel, not the air.** Check the fuel grade against the temperature, the filter and water separator for wax, and the lines for gelling. Winter-grade or blended diesel has a lower cloud and pour point than summer grade; below the cloud point wax crystals form and pack the filter media, so the engine cranks, builds compression and gets heated air but never receives enough fuel. Confirm with a restriction reading or by warming and replacing the filter, then look for air drawn in on the suction side. Injector return volume, valve timing and ECM calibration are all real checks, but none of them explains a fault that appears only when the temperature drops.

**Q37. Answer: D**

**Rated RPM is the full-load governed speed — the RPM at which the engine makes its advertised maximum horsepower.** Diesel torque peaks lower, around 1,100–1,500 RPM on many truck engines, while horsepower peaks at rated speed, around 1,800–2,100 RPM. Rated speed is not the engine's top speed: with the load removed, the governor lets the engine run up to high idle, the no-load governed speed, which sits above rated speed, and the over-speed limit sits above that again. Operating at rated RPM under load represents maximum power output from the engine.

**Q38. Answer: A**

**Relay valve: speeds up brake application at remote axles.** Without a relay valve, air must travel the full length of the vehicle (foot valve to rear brakes). The relay valve uses a small pilot signal from the foot valve to open a local valve, delivering air from a nearby reservoir rather than the foot valve — much faster response.

**Q39. Answer: B**

**Two-way check valve: passes the higher of two inlet pressures to its outlet.** Commonly used in spring brake circuits: either the service brake signal OR the parking brake supply can apply the spring brakes. The two-way check valve ensures that whichever signal is higher (emergency application OR service application) is what moves the spring brake piston. This allows the service brakes and parking brakes to share the same spring brake chambers without interfering with each other.

**Q40. Answer: C**

**Zero free travel leaves the release bearing touching the pressure plate, which bleeds off clamping force.** Even slight continuous contact takes load off the friction facings, and the clutch slips under torque. It also spins the release bearing all day under load, which is what kills the bearing. Free travel — the pedal movement before the release bearing reaches the release fingers — exists so that the pressure plate is fully clamped whenever the pedal is up. It is set to the figure in the manufacturer's service information and measured at the pedal pad, and it closes up on its own as the friction facings wear, which is why it is a scheduled check rather than a one-time adjustment. The release bearing to clutch brake clearance is a separate adjustment with its own figure and is set in the same procedure.

**Q41. Answer: C**

**Trip inspection report retention: at least 6 months.** NSC Standard 13 requires carriers to retain the original copy of each vehicle inspection report and certification of repairs for at least 6 months from the date the report was prepared. The driver forwards the original to the carrier within 20 calendar days of completing it, must be in possession of the required inspection report while driving, and must produce it on demand of an inspector. Defects recorded on a report must be corrected before the next required inspection, or within the timeframe the jurisdiction of travel specifies, and a vehicle with a major defect must not be driven at all.

**Q42. Answer: A**

**Zero oil pressure while running = engine emergency — shut down immediately.** Normal oil pressure at operating temperature is 25–65 PSI (varies by engine). Zero pressure means no oil is circulating — pump failure, oil loss, or a blocked oil pickup — and engine bearing damage begins within seconds without oil pressure. Treat zero oil pressure as an absolute emergency — never continue operating.

**Q43. Answer: C**

**Bobtail = drive axle nearly unloaded, so the system trims rear brake pressure to match.** With no trailer, a tractor's drive axle carries a small fraction of the weight it was braked for, and full rear application pressure would slide the wheels on all but the driest pavement. The bobtail proportioning system senses the no-trailer condition and delivers reduced pressure to the drive-axle chambers during service applications, keeping brake force in proportion to the load on the axle; normal delivery returns when a trailer is coupled. It does not shut the rear brakes off, nothing raises delivery above what the treadle meters, and trailer-circuit air is not redirected anywhere. The spring brake (park and emergency) function is mechanical and unaffected. Diagnostic angle: low delivery pressure at the drive axle on a bobtail tractor can be this system working as designed - confirm the proportioning function before condemning foundation parts for 'weak rear brakes'.

**Q44. Answer: A**

**A broken power spring cannot develop the clamping force the parking brake depends on.** A spring brake applies mechanically: exhausting the hold-off air lets a compressed coil spring drive the pushrod out. A fatigued, cracked or broken spring still applies the brake, but with too little force to hold the vehicle — and a grade is what exposes it. The stem rules out the usual rivals: full reservoirs mean supply is fine, and in-limit stroke means the foundation brake is adjusted. A relay valve that failed to release service air would leave the brakes dragging rather than free, closing the tractor protection valve applies the trailer spring brakes rather than releasing them, and the low pressure warning switch only drives a lamp and buzzer — nothing signals a spring brake to apply, because the application is what happens when air is taken away.

**Q45. Answer: D**

**Free play has no single limit — it changes with steering wheel diameter and with manual versus power steering.** National Safety Code Standard 11 Part B, Section 4 — Steering, sets the Canadian figures, and each province adopts them in its own inspection regulation. Measure at the rim with the engine running for power assist, turning the wheel each way until the front wheels just begin to move. The vehicle is rejected when lash exceeds 75 mm for a power-steering wheel of 500 mm or less, or 87 mm for one over 500 mm; manual steering is rejected past 87 mm and 100 mm. A second, higher set of figures in the same item marks a hazardous condition. So 4 inches (about 102 mm) is past every power-steering reject limit, but no truck can be judged until the wheel is measured and the steering type noted.

**Q46. Answer: D**

**A friction modifier.** A limited-slip differential adds a clutch pack or a cone clutch between the side gears and the differential case, so that when one wheel starts to spin some torque is still carried across to the wheel that still has grip. Those clutches are meant to slip a little and grip a little every time the axle goes round a corner, because the two wheels are travelling different distances and the differential has to let them. Plain hypoid gear oil makes them grab and release in steps rather than slipping smoothly, and the axle chatters and shudders through slow tight turns while the clutch faces glaze. The friction modifier is an additive that narrows the gap between static and sliding friction in the clutch pack so the slip is smooth and quiet. Some gear oils are sold already modified for limited-slip service; otherwise the modifier is added separately in the quantity the axle maker states. The other three answers describe choices made for other reasons: viscosity and the gear-oil rating are chosen for the ring and pinion and do not change because the differential is limited-slip, and a detergent package belongs to engine oil, which has combustion by-products to carry, not to axle oil.

**Q47. Answer: C**

**Battery discharge through "off" switch = switch not isolating.** Common causes: corroded or worn contacts in the isolator switch that no longer make a complete break, or a bypass circuit that bypasses the switch (alarm system, GPS tracker, fridge). Test with a clamp-on ammeter on the cable between switch and battery with switch OFF.

**Q48. Answer: D**

**Quick-release valve: rapid local air exhaust from brake chambers on release, allowing spring return and reducing brake drag.** Without a quick-release valve, air would have to travel all the way back through the treadle valve and exhaust at the driver's feet — taking too long and causing brake drag. The QR valve is located close to the brake chambers: when the pedal is released and line pressure drops, it immediately exhausts air locally at the chamber — rapid piston return and immediate brake release.

**Q49. Answer: D**

**Never increase fuse rating as a fix.** The fuse protects the wiring, not the component. A blown fuse means the circuit is drawing more current than the wire can safely carry. Increasing the rating allows wiring to overheat and potentially start a fire. Diagnose why the circuit draws excess current — short circuit, overloaded component, damaged wiring.

**Q50. Answer: A**

**Oil cooler bypass valve: opens on cold thick oil.** Cold engine oil has high viscosity and creates a high pressure differential across the oil cooler. The bypass valve opens to route oil around the cooler, ensuring adequate lubrication flow during cold starts before oil warms and thins. Once warm, the valve closes and oil flows through the cooler normally.

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